Ch 10

population :

sample :

random sample :

parameter :

statistic :

statistical inferenence :

sampling distribution :

$\{X_1, \cdots, X_n\}$ is a random sample from $f(x)$:

$E(X_i) = \mu$ $\qquad$ $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$

$Var(X_i) = \sigma^2$ $\qquad$ $E(\bar{X}) = \mu$

$Var(\bar{X}) = \frac{\sigma^2}{n}$

standard error of the sample mean : $SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{S}{\sqrt{n}}$

sample variance : $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$

$E(S^2) = \sigma^2$

$M_{\bar{X}}(t) = \left[M_X\left(\frac{t}{n}\right)\right]^n$

For $X_i \underbrace{\quad}_{} $ i.i.d. $N(\mu, \sigma^2)$

$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ ← (使用mgf得到dist.中参数!)

$\bar{X}$ and S^2 are independent

E.g. 10.8

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

E.g. 10.9

statistic $\underset{\substack{\downarrow \\ \text{(also a r.v.)}}}{T_n}$: any func. of a $\underset{\substack{\downarrow \\ \text{(a set of r.v.'s from pop.)}}}{\underline{\text{sample}}}$

sampling distribution

To find $Pr(T_n \le t)$ ☆☆☆ :

① 找 mgf

② Large sample approximation

Large sample theory

seq. of r.v.s. $Y_1, \cdots, Y_n$

usually, Y_i's statistics are calculated from a sample with size n

How to define limit of the seq.?

limit of the seq. of statistic is used to approximate the sampling dist. of sta.

(?) Simple Random sample

(?) $E(\bar{x})=\mu$, $Var(\bar{X})=\left(\frac{N-n}{N-1}\right)\frac{\sigma^2}{n}$

(未完)

(To E.g. 11.2, Ch 11)

X_n is converge in dist. to X

$\Leftrightarrow \lim_{n\to\infty} F_{X_n}(x) = F_X(x), \forall x$

$\Leftrightarrow X_n \xrightarrow{d} X$

(/or $X_n \xrightarrow{c} X$)

$\Uparrow$

X_n is converge in prob. to X

$\Leftrightarrow \forall \varepsilon, \lim_{n\to\infty} P_r(|X_n - X| \geq \varepsilon) = 0$

$\Leftrightarrow X_n \xrightarrow{p} X$

$\downarrow$

(化为 $F_{X_n}(\sim)$?)

$\Uparrow$

X_n is converge almost surely to X

$\Leftrightarrow P_r(\lim_{n\to\infty} X_n = X) = 1$

$\Leftrightarrow X_n \xrightarrow{a.s.} X$

$\Leftrightarrow \lim_{n\to\infty} X_n(\omega) = X(\omega)$,

$\forall \omega \in \Omega$ and $\omega \notin S$,

where $S \subset \Omega$ and $P_r(S) = 0$

(LLN: Law of Large Numbers)

WLLN (Weak LLN):

$$\overline{X}_n \xrightarrow{p} \mu = E(X_i)$$

(By Chebyshev's inequality,

$$Pr(|\overline{X}_n - \mu| \geqq \varepsilon) \leq \boxed{\frac{\sigma^2}{n\varepsilon^2}} \longrightarrow [illegible]$$

$$\left(Pr(|X - \mu| \geqq k\sigma) \leq \frac{1}{k^2} \right)$$

i.e., $\lim_{n\to\infty} Pr(|\overline{X}_n - \mu| \geqq \varepsilon) = 0$

SLLN (Strong LLN):

$$\overline{X}_n \xrightarrow{a.s.} \mu = E(X_i)$$

i.e., $(\lim_{n\to\infty} \overline{X}_n = \mu) = 1$

([illegible] More general ver.:

$$\Longleftrightarrow \begin{cases} E(|X_i|) < \infty \\ \text{(and)} \\ E(X_i) = \mu \end{cases} \Bigg)$$

CLT (Central Limit Theorem):

$$\mu = E(X_i), \quad \sigma^2 = Var(X_i)$$

$$\frac{\overline{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$$

as $n \to \infty$

i.e., $\lim_{n\to\infty} Pr\left(\frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \leqq x\right) = \Phi(x)$,

$$\forall \underbrace{-\infty <}_{\star} x \underbrace{< \infty}_{\circledast}$$

E.g.: Indep. Bernouli trials

$X_1, \cdots, X_n \overset{i.i.d.}{\sim} Ber(\frac{1}{2})$

$E(X_i) = \frac{1}{2}$, $Var(X_i) = \frac{1}{4}$

Show the value of n s.t. $Pr(\overline{X}_n > 0.06) \leq 0.025$

CLT: $\frac{\overline{X}_n - \frac{1}{2}}{\frac{\sqrt{\frac{1}{4}}}{\sqrt{n}}} \xrightarrow{d} N(0, 1)$

$$Pr(\overline{X}_n > 0.6)$$

⊛ $$= Pr\left(\frac{\overline{X}_n - \frac{1}{2}}{\frac{\sqrt{\frac{1}{4}}}{\sqrt{n}}} > \frac{0.6 - \frac{1}{2}}{\frac{\sqrt{\frac{1}{4}}}{\sqrt{n}}}\right)$$

$$= Pr\left(Z > \frac{0.2}{\textcircled{?}}\sqrt{n}\right)$$

$$= 1 - Pr(Z \leq 0.2\sqrt{n})$$

$$= 1 - \Phi(0.2\sqrt{n}) \leq 0.025$$

$$0.2\sqrt{n} \geq \Phi^{-1}(0.975) = 1.96 \ \underline{\textcircled{?}}$$

$$n \geq \left(\frac{1.96}{0.2}\right)^2 = 96.04$$

$$n \in \mathbb{Z} \Rightarrow n = 97$$